might be the same nine particles accompanied by any number of chargeless particles (such as photons). As was the case with the hidden sequence of letters when we only had the clue about the three y's, the possibilities for the contents of the box are endless.

But let's now imagine that, as in the case of the linguistic puzzle, we are given two further clues: The theory describing the world—and hence the contents of the box—is supersymmetric, and the content of the box has the minimum mass consistent with the first clue of having three units of charge. Based on the insights of E. Bogomoln'yi, Manoj Prasad, and Charles Sommerfeld, physicists have shown that this specification of a tight organizational framework (the framework of supersymmetry, the analog of the English language) and a "minimality constraint" (minimum mass for a chosen amount of electric charge, the analog of a minimum word length for a chosen number of y's) implies that the identity of the hidden contents is nailed down uniquely. That is, merely by ensuring that the contents of the box is the lightest it could possibly be and still have the specified charge, physicists showed that its identity is fully established. Constituents of minimum mass for a chosen value of charge are known as BPS states, in honor of their three discoverers.8

The important thing about BPS states is that their properties are uniquely, easily, and exactly determined without resort to a perturbative calculation. This is true regardless of the value of the coupling constants. That is, even if the string coupling constant is large, implying that the perturbative approach is invalid, we are still able to deduce the exact properties of the BPS configurations. The properties are often called nonperturbative masses and charges since their values transcend the perturbative approximation scheme. For this reason, you can also think of BPS as standing for "beyond perturbative states."

The BPS properties exhaust only a small part of the full physics of a chosen string theory when its coupling constant is large, but they nonetheless give us a tangible grip on some of its strong coupling characteristics. As the coupling constant in a chosen string theory is increased beyond the realm accessible to perturbation theory, we anchor our limited understanding in the BPS states. Like a few choice words in a foreign tongue, we will find that they will take us quite far.